

嵌入式作業系統-LAB2

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1. 移除不必要的功能, 縮小 RPi OS 的 image size (應說明: 目的、移除哪些功能、如何移除、前後 image size 與功能的比較)

目的:

為了測試多種 config 對於 image size 與效能的影響, 我並沒有針對特定應用場景做針對性方向的修改, 而是為了測試各功能對於 image size 與效能的影響。

移除前效能:

```
shihcw@raspberrypi:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second:   272.79

General statistics:
  total time:          10.0020s
  total number of events: 2730

Latency (ms):
  min:                 3.63
  avg:                 3.66
  max:                 5.33
  95th percentile:    3.68
  sum:                 9998.79

Threads fairness:
  events (avg/stddev): 2730.0000/0.00
  execution time (avg/stddev): 9.9988/0.00
```

原效能:CPU 效能(透過質數運算測試)

```

shihcw@raspberrypi:~$ sysbench memory --memory-total-size=1G run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Running memory speed test with the following options:
  block size: 1KiB
  total size: 1024MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 1048576 (1222808.51 per second)

1024.00 MiB transferred (1194.15 MiB/sec)

General statistics:
  total time:                0.8516s
  total number of events:    1048576

Latency (ms):
  min:                       0.00
  avg:                       0.00
  max:                       5.33
  95th percentile:         0.00
  sum:                       428.04

Threads fairness:
  events (avg/stddev):       1048576.0000/0.00
  execution time (avg/stddev): 0.4280/0.00

```

原效能: 記憶體讀寫速度

```

shihcw@raspberrypi:~$ sysbench --test=fileio --file-total-size=1G --file-test-mode=rndrw run
WARNING: the --test option is deprecated. You can pass a script name or path on the command line without any options.
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Extra file open flags: (none)
128 files, 8MiB each
1024 total file size
Block size 16KiB
Number of IO requests: 0
Read/write ratio for combined random IO test: 1.50
Periodic fsync enabled, calling fsync() each 100 requests.
Calling fsync() at the end of test, Enabled.
Using synchronous I/O mode
Doing random r/w test
Initializing worker threads...

Threads started!

File operations:
  reads/s:                   271.29
  writes/s:                  180.83
  fsyncs/s:                  582.12

Throughput:
  read, MiB/s:               4.24
  written, MiB/s:            2.83

General statistics:
  total time:                10.1078s
  total number of events:    10333

Latency (ms):
  min:                       0.00
  avg:                       0.97
  max:                       69.23
  95th percentile:         4.33
  sum:                       9979.18

Threads fairness:
  events (avg/stddev):       10333.0000/0.00
  execution time (avg/stddev): 9.9792/0.00

```

原效能: 磁碟 I/O 讀寫性能



原始 Image_size

移除的功能(將會標記測試次數 Test_n):

● (Test_1) ——Device Drivers ——

[] PCI support

< > PCCard (PCMCIA/CardBus) support

< > Sound card support

< > Serial ATA and Parallel ATA drivers (libata)

[] Multiple devices driver support (RAID and LVM)

	原效能	Test_1
Image size	23.6MB	23.6MB
Execution time (avg/stddev) CPU 效能 (質數運算)	9.9988	9.9980
Execution time (avg/stddev) 記憶體讀寫速度	0.4280	0.4007
Execution time (avg/stddev) 隨機讀寫磁碟效能	9.9792	9.9763

```

shihcw@raspberrypi:~ $ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second:   272.12

General statistics:
  total time:          10.0011s
  total number of events: 2723

Latency (ms):
  min:                 3.64
  avg:                 3.67
  max:                 6.54
  95th percentile:    3.68
  sum:                 9998.03

Threads fairness:
  events (avg/stddev): 2723.0000/0.00
  execution time (avg/stddev): 9.9980/0.00

```

Test_1:CPU 效能(透過質數運算測試)

```

shihcw@raspberrypi:~ $ sysbench memory --memory-total-size=1G run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Running memory speed test with the following options:
  block size: 1KiB
  total size: 1024MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 1048576 (1314532.83 per second)

1024.00 MiB transferred (1283.72 MiB/sec)

General statistics:
  total time:          0.7921s
  total number of events: 1048576

Latency (ms):
  min:                 0.00
  avg:                 0.00
  max:                 0.01
  95th percentile:    0.00
  sum:                 400.72

Threads fairness:
  events (avg/stddev): 1048576.0000/0.00
  execution time (avg/stddev): 0.4007/0.00

```

Test_1:記憶體讀寫速度

```

shihcw@raspberrypi:~$ sysbench --test=fileio --file-total-size=1G --file-test-mode=rndrw run
WARNING: the --test option is deprecated. You can pass a script name or path on the command line without any options.
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Extra file open flags: (none)
128 files, 8MiB each
1GiB total file size
Block size 16KiB
Number of IO requests: 0
Read/Write ratio for combined random IO test: 1.50
Periodic fsync enabled, calling fsync() each 100 requests.
Calling fsync() at the end of test, Enabled.
Using synchronous I/O mode
Doing random r/w test
Initializing worker threads...

Threads started!


File operations:
  reads/s:          291.74
  writes/s:         194.49
  fsyncs/s:         629.63

Throughput:
  read, MiB/s:      4.56
  written, MiB/s:    3.04

General statistics:
  total time:        10.0672s
  total number of events: 11117

Latency (ms):
  min:               0.00
  avg:               0.90
  max:               34.60
  95th percentile:  4.41
  sum:               9976.30

Threads fairness:
  events (avg/stddev): 11117.0000/0.00
  execution time (avg/stddev): 9.9763/0.00

```

Test_1:磁碟 I/O 讀寫性能

功能比較與觀察:

- [1]CPU 性能：關閉這些功能後，CPU 測試的每秒處理事件數基本持平（從 272.79 下降到 272.12），總計處理的事件數量幾乎不變，執行時間維持在 10 秒左右。這表示這些驅動的移除對於 CPU 密集型計算幾乎沒有影響。
- [2]記憶體性能：記憶體測試顯示總傳輸速率略有上升（1194.15 MiB/sec → 1283.72 MiB/sec），總操作次數也增加，表明關閉部分驅動後，可能釋放了一些系統資源，使得記憶體操作的效率有所提升。
- [3]I/O 性能：I/O 測試顯示讀寫速度都有提升（讀取 4.24 MiB/s → 4.56 MiB/s，寫入 2.83 MiB/s → 3.04 MiB/s），fsync 次數也增加（582.12/s → 629.63/s），表明 I/O 效能獲得了一定的增強。這可能是由於 SATA/PATA 相關驅動被禁用，使系統避免了不必要的磁碟管理開銷。

內核裁減（關閉不必要的驅動）對 CPU 幾乎無影響，但對記憶體和 I/O 性能有輕微的優化效果，特別是 I/O 效能提升較為明顯。這表明減少系統負擔的確有助於提升效能，尤其是在嵌入式系統或資源受限的環境下，這樣的內核優化可能帶來更顯著的效益。

● (Test_2) ——Networking support ——

- [] Amateur Radio support
- < > CAN bus subsystem support
- < > RF switch subsystem support
- < > Plan 9 Resource Sharing Support (9P2000)

	Test_1	Test_2
Image size	23.6MB	23.6MB
Execution time (avg/stddev) CPU 效能（質數運算）	9.9980	9.9978
Execution time (avg/stddev) 記憶體讀寫速度	0.4007	0.4004
Execution time (avg/stddev) 隨機讀寫磁碟效能	9.9763	9.9744

```
shihcw@raspberrypi:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second: 272.63

General statistics:
  total time:          10.0007s
  total number of events: 2728

Latency (ms):
  min:                 3.63
  avg:                 3.66
  max:                 5.75
  95th percentile:    3.68
  sum:                 9997.77

Threads fairness:
  events (avg/stddev): 2728.0000/0.00
  execution time (avg/stddev): 9.9978/0.00
```

Test_2:CPU 效能(透過質數運算測試)

```

shihcw@raspberrypi:~$ sysbench memory --memory-total-size=1G run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Running memory speed test with the following options:
  block size: 1KiB
  total size: 1024MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 1048576 (1298784.21 per second)

1024.00 MiB transferred (1268.34 MiB/sec)

General statistics:
  total time:                0.8017s
  total number of events:    1048576

Latency (ms):
  min:                       0.00
  avg:                       0.00
  max:                       0.02
  95th percentile:          0.00
  sum:                       400.38

Threads fairness:
  events (avg/stddev):       1048576.0000/0.00
  execution time (avg/stddev): 0.4004/0.00

```

Test_2: 記憶體讀寫速度

```

shihcw@raspberrypi:~$ sysbench --test=fileio --file-total-size=1G --file-test-mode=rndrw run
WARNING: the --test option is deprecated. You can pass a script name or path on the command line without any options.
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Extra file open flags: (none)
128 files, 8MiB each
1GiB total file size
Block size 16KiB
Number of IO requests: 0
Read/Write ratio for combined random IO test: 1.50
Periodic fsync enabled, calling fsync() each 100 requests.
Calling fsync() at the end of test, Enabled.
Using synchronous I/O mode
Doing random r/w test
Initializing worker threads...

Threads started!

File operations:
  reads/s:                298.43
  writes/s:               198.95
  fsyncs/s:               646.40

Throughput:
  read, MiB/s:            4.66
  written, MiB/s:         3.11

General statistics:
  total time:              10.0436s
  total number of events:  11370

Latency (ms):
  min:                     8.80
  avg:                     8.88
  max:                     32.94
  95th percentile:        4.41
  sum:                     9974.42

Threads fairness:
  events (avg/stddev):     11370.0000/0.00
  execution time (avg/stddev): 9.9744/0.00

```

Test_2: 磁碟 I/O 讀寫性能

功能比較與觀察：

從 sysbench 測試結果來看，關閉 Networking support (Test_2) 中的某些功能對 CPU 和 Memory 性能幾乎沒有影響，而 I/O 讀寫性能則稍有提升。這可能是因為關閉某些網路相關的子系統減少了中斷和驅動層的開銷，從而提升了 I/O 效率，但對計算和記憶體存取影響不大。

● (Test_3) ——General setup——

Compiler optimization level (Optimize for performance (-O2))改成
Compiler optimization level (Optimize for size (-Os))

——Device Drivers ——

[*] PCI support

(另外將 PCI support 改回，改了會讓 image 大小不變)

	Test_2	Test_3
Image size	23.6MB	21.1MB
Execution time (avg/stddev) CPU 效能（質數運算）	9.9978	9.9970
Execution time (avg/stddev) 記憶體讀寫速度	0.4004	0.4019
Execution time (avg/stddev) 隨機讀寫磁碟效能	9.9744	9.9742



Test_3:Image_size

```

shihcw@raspberrypi:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second:   273.03

General statistics:
  total time:          10.0005s
  total number of events: 2732

Latency (ms):
  min:                 3.64
  avg:                 3.66
  max:                 5.33
  95th percentile:    3.68
  sum:                 9996.98

Threads fairness:
  events (avg/stddev): 2732.0000/0.00
  execution time (avg/stddev): 9.9970/0.00

```

Test_3:CPU 效能(透過質數運算測試)

```

shihcw@raspberrypi:~$ sysbench memory --memory-total-size=1G run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Running memory speed test with the following options:
  block size: 1KiB
  total size: 1024MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 1048576 (1299984.80 per second)

1024.00 MiB transferred (1269.52 MiB/sec)

General statistics:
  total time:          0.8007s
  total number of events: 1048576

Latency (ms):
  min:                 0.00
  avg:                 0.00
  max:                 0.02
  95th percentile:    0.00
  sum:                 401.88

Threads fairness:
  events (avg/stddev): 1048576.0000/0.00
  execution time (avg/stddev): 0.4019/0.00

```

Test_3:記憶體讀寫速度

```
mtk@mtk-raspberrypi: ~ $ sysbench --test=fileio --file-total-size=1G --file-test-mode=rndrw run
WARNING: the --test option is deprecated. You can pass a script name or path on the command line without any options.
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time

Extra file open flags: (none)
128 files, 8MiB each
1GiB total file size
Block size 16KiB
Number of IO requests: 0
Read/write ratio for combined random IO test: 1.50
Periodic fsync enabled, calling fsync() each 100 requests.
Calling fsync() at the end of test, Enabled.
Using synchronous I/O mode
Doing random r/w test
Initializing worker threads...

Threads started!

File operations:
  reads/s:          291.55
  writes/s:         194.36
  fsyncs/s:         628.31

Throughput:
  read, MiB/s:      4.56
  written, MiB/s:    3.04

General statistics:
  total time:        10.0747s
  total number of events: 11198

Latency (ms):
  min:               0.00
  avg:               0.90
  max:               31.68
  95th percentile:  4.49
  sum:               9974.17

Threads fairness:
  events (avg/stddev): 11198.0000/0.00
  execution time (avg/stddev): 9.9742/0.00
```

Test_3:磁碟 I/O 讀寫性能

功能比較與觀察：

這次的修改主要是針對 減少系統映像大小，但對於 運算效能影響不大，可能對 I/O 效能有小幅影響。

- (Test_4) ——File systems——
 - < > Reiserfs support (deprecated)
 - < > JFS filesystem support
 - < > XFS filesystem support
 - < > GFS2 file system support
 - < > OCFS2 file system support

	Test_3	Test_4
Image size	21.1MB	21.1MB
Execution time (avg/stddev) CPU 效能（質數運算）	9.9970	9.9976
Execution time (avg/stddev) 記憶體讀寫速度	0.4019	0.4019
Execution time (avg/stddev) 隨機讀寫磁碟效能	9.9742	9.9811

```

shihcw@raspberrypi:~$ sysbench cpu --cpu-max-prime=20000 run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Prime numbers limit: 20000

Initializing worker threads...

Threads started!

CPU speed:
  events per second:   272.30

General statistics:
  total time:          10.0013s
  total number of events: 2725

Latency (ms):
  min:                 3.64
  avg:                 3.67
  max:                 6.61
  95th percentile:    3.68
  sum:                 9997.65

Threads fairness:
  events (avg/stddev): 2725.0000/0.00
  execution time (avg/stddev): 9.9976/0.00

```

Test_4:CPU 效能(透過質數運算測試)

```

shihcw@raspberrypi:~$ sysbench memory --memory-total-size=1G run
sysbench 1.0.20 (using system LuaJIT 2.1.0-beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Running memory speed test with the following options:
  block size: 1KiB
  total size: 1024MiB
  operation: write
  scope: global

Initializing worker threads...

Threads started!

Total operations: 1048576 (1306972.84 per second)

1024.00 MiB transferred (1276.34 MiB/sec)

General statistics:
  total time:          0.7965s
  total number of events: 1048576

Latency (ms):
  min:                 0.00
  avg:                 0.00
  max:                 0.02
  95th percentile:    0.00
  sum:                 401.88

Threads fairness:
  events (avg/stddev): 1048576.0000/0.00
  execution time (avg/stddev): 0.4019/0.00

```

Test_4:記憶體讀寫速度

```

shihua@raspberrypi:~$ sysbench --test=fileio --file-total-size=1G --file-test-mode=rndrw run
WARNING: the --test option is deprecated. You can pass a script name or path on the command line without any options.
sysbench 1.0.20 (using system LuaJIT 2.1.0-Beta3)

Running the test with following options:
Number of threads: 1
Initializing random number generator from current time


Extra file open flags: (none)
128 files, 8MiB each
1GiB total file size
Block size 16KiB
Number of IO requests: 0
Read/Write ratio for combined random IO test: 1.50
Periodic fsync enabled, calling fsync() each 100 requests.
Calling fsync() at the end of test, enabled.
Using synchronous I/O mode
Doing random r/w test
Initializing worker threads...

Threads started!


File operations:
  reads/s:          279.72
  writes/s:         186.48
  fsyncs/s:          599.90


Throughput:
  read, MiB/s:       4.37
  written, MiB/s:    2.91


General statistics:
  total time:        10.0762s
  total number of events: 10620


Latency (ms):
  min:               0.00
  avg:               0.94
  max:               67.03
  95th percentile:  4.41
  sum:               9981.05


Threads fairness:
  events (avg/stddev): 10620.0000/0.00
  execution time (avg/stddev): 9.9811/0.00

```

Test_4:磁碟 I/O 讀寫性能

功能比較與觀察：

[1]CPU 性能：

- CPU 效能幾乎沒有變化，執行時間、事件數量和延遲數據都保持在相同範圍內。
- 這表明這些文件系統的支持與 CPU 密集型計算（如素數計算）幾乎無關。

[2]記憶體性能：

- 內存吞吐量（MiB/sec）有小幅提升。
- 這可能是因為關閉不必要的文件系統驅動，減少了內核內存佔用，導致內存管理更高效。

[3]I/O 性能：

- 隨機讀寫（reads/s, writes/s）以及吞吐量（MiB/s）均有小幅提升。
- 關閉這些不必要的文件系統，可能減少了內核在 I/O 處理時的額外開銷，提升了 I/O 效能。

關閉不常用的文件系統驅動，對 CPU 性能影響不大，但 內存和 I/O 性能略有提升。這說明你的優化策略在 減少內核開銷 方面是有效的，特別是對於存儲性能的影響較為正面。如果你的系統不需要這些文件系統支持，那麼這樣的精簡優化是值得保留的。

2. 在網路上下載可以評測 kernel 的套件，比較原本與更改後的 kernel（應說明：下載哪一套件、改善哪方面的效能與評測的結果）

- 下載 Sysbench 套件

Sysbench 可以測試 CPU、記憶體和磁碟 I/O。

- CPU 測試：

指令：`sysbench cpu --cpu-max-prime=20000 run`

指令說明：sysbench 會產生隨機數作為測試種子。使用一個 單執行緒（1 thread）來計算 2 到 20000 之間的所有質數。

測試結束後，報告：

events per second → 越高越好，代表 CPU 運算能力越強。

total time → 測試運行的總時間。

latency → 單個計算事件所需的時間。

- 記憶體測試：

指令：`sysbench memory --memory-total-size=1G run`

指令說明：

預設會執行 1KiB 的記憶體塊寫入，直到達到 1GB 的總寫入量。

sysbench 會記錄：

Total operations → 總共執行多少次記憶體操作。

Memory Throughput → MiB/sec，表示記憶體讀寫速度，數字越大越好。

Latency → 每次記憶體操作的延遲時間。

- 磁碟 I/O 測試

指令：`sysbench --test=fileio --file-total-size=1G --file-test-mode=rndrw run`

指令說明：

建立一個 1GB 的測試檔案（file-total-size=1G）。

隨機讀寫（rndrw = random read & write）來模擬實際應用場景。

執行 10 秒，並記錄 I/O 相關數據：

reads/s → 每秒讀取多少次，數字越大越好。

writes/s → 每秒寫入多少次，數字越大越好。

fsync/s → 表示磁碟的同步寫入速度。

MiB/sec → 讀取 & 寫入的實際速度，越高越好。

Latency avg → 每個 I/O 操作的平均延遲，越低越好。

- 移除的功能：

——Device Drivers ——

[] PCI support (造成 Image_size 不變的主因)

< > PCCard (PCMCIA/CardBus) support ----

< > Sound card support ----

< > Serial ATA and Parallel ATA drivers (libata) ----

[] Multiple devices driver support (RAID and LVM) ----

——Networking support ——

[] Amateur Radio support ----

< > CAN bus subsystem support --->

< > RF switch subsystem support ----

< > Plan 9 Resource Sharing Support (9P2000) ----

——General setup——

Compiler optimization level (Optimize for performance (-O2)) 改為

Compiler optimization level (Optimize for size (-Os))

——Device Drivers ——

[*] PCI support (另外將 PCI support 改回，改了會讓 image 大小不變)

——File systems——

< > Reiserfs support (deprecated)

< > JFS filesystem support

< > XFS filesystem support

< > GFS2 file system support

< > OCFS2 file system support

● 移除功能前後的效能比較：

	原效能	Test_4
Image size	23.6MB	21.1MB
Execution time (avg/stddev) CPU 效能 (質數運算)	9.9988	9.9976
Execution time (avg/stddev) 記憶體讀寫速度	0.4280	0.4019
Execution time (avg/stddev) 隨機讀寫磁碟效能	9.9792	9.9811

功能比較與觀察：

[1] Image 大小減少：

透過後來加回 PCI support 及其他功能的移除，使得 Image_size 從 23.6MB 減少至 21.1MB，代表精簡作業系統內核確實有效減少了不必要的組件。

[2] CPU 計算效能基本不變：

在 sysbench CPU 測試中，數值變動極小（從 9.9988 到 9.9976 秒），說明改變編譯優化（從 -O2 變為 -Os）對於 CPU 效能影響不大，可能因為 -Os 仍保留了一些效能最佳化特性。

[3] 記憶體讀寫效能稍有提升：

記憶體測試的執行時間從 0.4280 秒降低到 0.4019 秒，顯示記憶體傳輸效能提升，這可能是因為移除部分驅動降低了系統開銷，讓記憶體 I/O 更有效率。

[4] 隨機讀寫磁碟效能幾乎持平：

I/O 測試的執行時間變化微乎其微（9.9792 秒 vs. 9.9811 秒），顯示移除特定檔案系統支援對於標準磁碟 I/O 影響不大，這可能是因為你仍然保留了常用的檔案系統（如 EXT4）。

精簡化內核確實成功減少了 Image 的大小，同時在不影響 CPU 計算能力的前提下，微幅提升了記憶體 I/O 效能，且對隨機磁碟 I/O 影響不大。內核裁減對於內存效率有些許優化效果，而不影響關鍵運算效能。

3. 使 kernel 支援 real-time 功能，可參照講義的作法，依照自己 kernel 的版本來 patch（應說明：欲下載哪一版本的補丁、如何下載與下載的結果）

首先透過講義提供的指令確認版本：

```
shihcw@raspberrypi:~ $ uname -r
6.1.93-v8+
```

```
shihcw@shihcw-GF65-Thin-9SD:~/linux$ head Makefile -n 4
# SPDX-License-Identifier: GPL-2.0
VERSION = 6
PATCHLEVEL = 1
SUBLEVEL = 93
```

確定好版本後根據助教所提供的網址下載對應的 6.1 且含有 realtime 的相關補丁：

我這邊選擇的是 patch-6.1.128-rt49.patch.gz

Index of /pub/linux/kernel/projects/rt/6.1/

../	20-Feb-2023 16:49	-
incr/	11-Feb-2025 21:01	-
older/	11-Feb-2025 21:01	62K
patch-6.1.128-rt49.patch.gz	11-Feb-2025 21:01	833
patch-6.1.128-rt49.patch.sign	11-Feb-2025 21:01	53K
patch-6.1.128-rt49.patch.xz	11-Feb-2025 21:01	100K
patches-6.1.128-rt49.tar.gz	11-Feb-2025 21:01	833
patches-6.1.128-rt49.tar.sign	11-Feb-2025 21:01	78K
patches-6.1.128-rt49.tar.xz	11-Feb-2025 21:05	1261
sha256sums.asc		

透過將講義上所給定的指令修改成自己所需的版本：

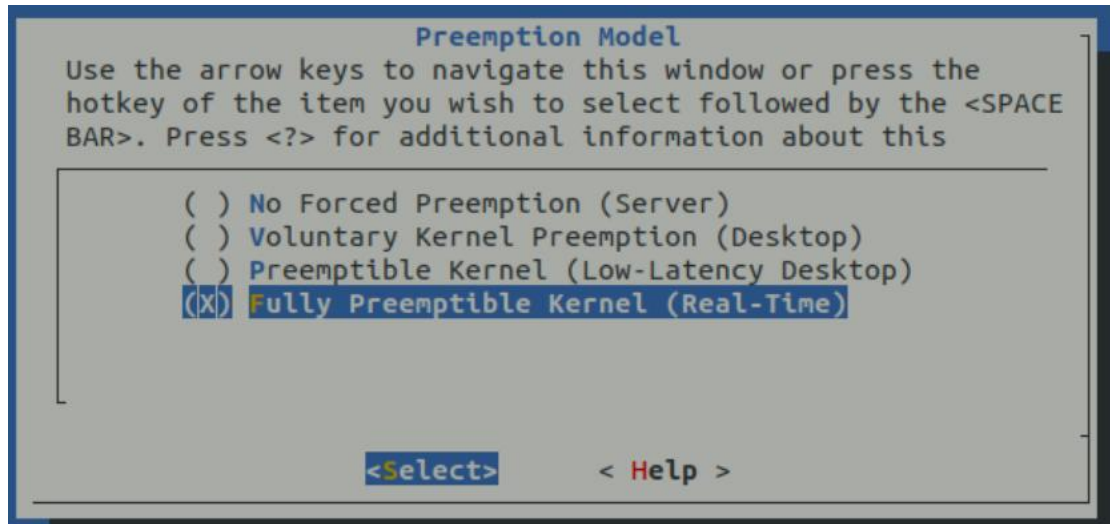
```
cd linux
```

```
wget https://www.kernel.org/pub/linux/kernel/projects/rt/6.1/older/
patch-6.1.128-rt49.patch.gz
```

```
gunzip patch-6.12.8-rt49.patch.gz
```

```
cat patch-6.1.128-rt49.patch | patch -p1
```

接著重複第五步驟開始,即可完成 realtime patch 的補丁下載。可以觀察到在 Preemption Model 中出現原先沒有的選項,Fully Preemptible Kernel(Real-Time)。



進入 RPi 後輸入 `uname -r`

```
shihcw@raspberrypi:~ $ uname -r  
6.1.93-rt49-v8+
```

可以看到版本成功的變為 realtime 的版本,這也表示補丁成功的使用！